



MANUAL TOOLS COMPANY

Manufacturer of Coke Oven Machinery



30+ YEARS OF
EXCELLENCE

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Dhanbad, Jharkhand | manualtoolsco.com



DOOR LIFTING POWER WINCH

COKE OVEN MAINTENANCE EQUIPMENT



KEY CAPABILITY:

A power winch for lifting heavy coke oven doors. Its self-locking worm reducer gearbox holds the load steady even during a power failure. 5 - 7.5 HP motor, up to 5 tons.

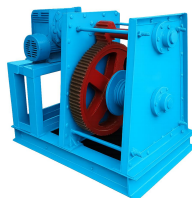
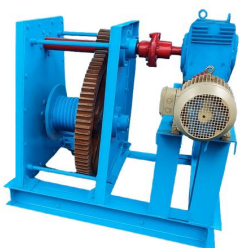
PERFORMANCE

- ✓ Lifting Capacity: 2.5 - 5 Tons
- ✓ Motor: 5 - 7.5 HP (3-Phase)
- ✓ Operation: Vertical Lift
- ✓ Lifting Speed: approx. 2 - 4 m/min

SAFETY & DESIGN

- ✓ Self-Locking Worm Reducer Gearbox
- ✓ Cast Steel / Phosphor Bronze Gears
- ✓ Grooved Steel Drum for Wire Rope
- ✓ Electro-Magnetic Motor Brake Available

PRODUCT GALLERY



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HOW IT WORKS

1. Drive Activation

The electric motor engages and drives the worm gearbox input shaft through a coupling.

2. Torque Multiplication

The worm shaft drives the worm wheel, cutting speed and multiplying torque for a smooth, non-jerky lift.

3. Vertical Lift

The output shaft turns the grooved drum, winding the wire rope and lifting the oven door to the required height.

APPLICATIONS

Coke Oven Doors	Lifting and positioning heavy oven doors during charging.
Heavy Dampers	Lifting isolation dampers in flue gas ducts in power plants and steel mills.
Maintenance Bays	General vertical lifting where overhead cranes are not available.

COMMON QUESTIONS

Q. Why a worm gearbox instead of helical?

A. Worm gearboxes are self-locking: if power fails, the door's weight cannot drive the motor backwards.

Q. What is the lifting speed?

A. Slow and controlled, approximately 2 - 4 metres/minute, for safety.

Q. Does it come with a brake?

A. We recommend and supply an electro-magnetic brake on the motor shaft for double safety.

Q. What capacity do I need for a coke oven?

A. Standard coke oven doors usually need 2.5 to 5 tons, depending on battery height and door weight.

Q. Is the wire rope included?

A. Yes, a standard length is included; rope length and diameter can be customised.

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